

Subject Code	ENGL5037
Subject Title	AI-Powered Language Analysis and Communication Strategies
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	None
Objectives	This subject provides an overview of generative AI and its potential applications as both a tool for language analysis and as a communicative tool itself. As such, the subject introduces students to the affordances and limitations of incorporating AI in analysing language in various professional contexts, including healthcare communication, language learning, translation, and advertising. In addition, the subject also looks at the many issues surrounding the usage of AI as a communicative and analytic tool, including its many affordances and limitations, reactions to AI-generated texts and images, perceptions of the efficacy of AI-generated and AI-adjacent analyses, and the role of bias and prevalent power structures in AI-generated responses.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Identify key affordances and limitations related to generative AI and its usage as an analytical tool. - Showcase proficiency in using generative AI as an analytical tool in various professional contexts. - Identify key affordances and limitations of generative AI as a content generator and as a communicative tool. - Showcase proficiency in using generative AI as a content-generating tool. - Understand the various forms of bias inherent in, and the relevant concerns with, AI-generated responses. - Use generative AI appropriate for creative problem-solving.
Subject Synopsis/ Indicative Syllabus	- Generative AI (GenAI) and human speech - GenAI, media creation, and multimodality - Bases and biases in GenAI models - Accuracy and reliability in responses - Biases in responses - GenAI as a tool naturally biased in favour of disinformation

	4. Using GenAI as an analytical tool in professional contexts a. Language learning b. Academic communication c. Healthcare communication d. Advertising e. Translation 5. Using AI as a content-generating communicative tool a. Reactions to AI-generated content b. Integration of human-generated and AI-generated content																																																					
Teaching/Learning Methodology	This subject adopts a hands-on approach to understanding the affordances and limitations of generative AI (GenAI) as an analytical tool and as a communicative tool. Students are expected to complete the course readings prior to class and to utilise GenAI tools in the classroom to recreate first-hand how said readings incorporated GenAI. Students are also expected to utilise GenAI in creating topic-appropriate examples to discuss each lecture. To promote active learning, students are expected to participate actively in each lecture and to find and bring in their own supplementary readings and AI-generated responses to the lecture topics. Through this, students should develop a deeper understanding of how GenAI has been used in the literature and an understanding of how it can be better utilised in the future. As such, students are expected to keep in mind the biases, strengths, and weaknesses inherent to GenAI and to work with and around these as much as possible when generating their own work for the class.																																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Individual participation and example sharing</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Individual presentation and analysis of AI-generated material</td><td>30</td><td></td><td></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Pair analysis of GenAI as an analytical tool</td><td>30</td><td>✓</td><td>✓</td><td></td><td></td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table>								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Individual participation and example sharing	40	✓	✓	✓	✓	✓	✓	2. Individual presentation and analysis of AI-generated material	30			✓	✓	✓	✓	3. Pair analysis of GenAI as an analytical tool	30	✓	✓			✓	✓	Total	100 %						
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	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Individual participation and example sharing (40%) Each lecture, students are expected to bring in their own AI-generated responses relevant to the topic of that lecture and to actively discuss said responses. Discussions should not be limited to the student's own examples but should also include discussions of other student's examples. These discussions should showcase understanding of that week's and prior weeks' readings. Students are expected to participate and share actively, without needing to be prompted by the subject lecturer to participate. Individual presentation and analysis of AI-generated material (30%) Using GenAI, students will generate communicative material (e.g., an advertisement) relevant to the topics given by the course lecturer. Students will then give a 10-minute presentation consisting of a 2-to-3-minute overview of the generated material and a 7-to-8-minute analysis of the generated material. The analysis should incorporate both the readings discussed in class and additional relevant academic readings. Pair analysis of GenAI as an analytical tool (30%) In pairs, students will present a real-world issue that has not been discussed in the literature, will demonstrate how GenAI can be used to analyse and address said issue (e.g., how GenAI can analyse social media responses and be used to improve business-to-consumer interactions), and will analyse said issue using GenAI. Presentations should be 10 minutes long and should consist of a 2-to-3-minute overview of the issue, a 5-to-6-minute explanation of the GenAI analysis, and a 2-minute explanation of the potential limitations and biases in the AI-generated response.	
Student Study Effort Expected	Class contact	
	▪ Seminars	39 Hrs.
	Other student study effort:	
	▪ Private study	81 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Ayers, J. W., Poliak, A., Dredze, M., Leas, E. C., Zhu, Z., Kelley, J. B., Faix, D. J., Goodman, A. M., Longhurst, C. A., Hogarth, M., & Smith, D. M. (2003). Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum. <i>JAMA Internal Medicine</i>, 183(6), 589–596. doi:10.1001/jamainternmed.2023.1838	

	Borovik, I., & Viro, V. (2023). Co-performing music with AI: Real-time performance control using speech and gestures. <i>HHAI 2023: Augmenting Human Intellect</i>, 368, 340-350. doi:10.3233/FAIA230097 Casal, J. E., & Kessler, M. (2023). Can linguists distinguish between ChatGPT/AI and human writing?: A study of research ethics and academic publishing. <i>Research Methods in Applied Linguistics</i>, 2(3), 100068. https://doi.org/10.1016/j.rmal.2023.100068. Desaire, H., Chua, A. E., Kim, M.-G., & Hua, D. (2023). Accurately detecting AI text when ChatGPT is told to write like a chemist. <i>Cell Reports Physical Science</i>, 4, 101672. https://doi.org/10.1016/j.xcrp.2023.101672 Diwanji, V. S., Lee, J., & Cortese, J. (2022). Deconstructing the role of artificial intelligence in programmatic advertising: At the intersection of automation and transparency. <i>Journal of Strategic Marketing</i>. doi:10.1080/0965254X.2022.2148269 Goodman, R. S., Patrinely, J. R., Stone, C. A., et al. (2023). Accuracy and reliability of chatbot responses to physician questions. <i>JAMA Network Open</i>, 6(10), e2336483. doi:10.1001/jamanetworkopen.2023.36483 Gorichanaz, T. (2023). Accused: How students respond to allegations of using ChatGPT on assessments. <i>Learning: Research and Practice</i>, 9(2), 183-196. DOI: 10.1080/23735082.2023.2254787 Grobys, K., Kolari, J. W., & Niang, J. (2022). Man versus machine: On artificial intelligence and hedge funds performance. <i>Applied Economics</i>, 54(40), 4632-4646. doi:10.1080/00036846.2022.2032585 Gutherz, G., Gordin, S., Sáenz, L., Levy, O., & Berant, J. (2023). Translating Akkadian to English with neural machine translation. <i>PNAS Nexus</i>, 2(5), 1-10. https://doi.org/10.1093/pnasnexus/pgad096 Hessel, J., Marasovic, A., Hwang, J. D., Lee, L., Da, J., Zellers, R., Mankoff, R., & Choi, Y. (2023). Do androids laugh at electric sheep? Humor “understanding” benchmarks from The New Yorker caption contest. <i>Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics</i>, Canada, 1, 688–714. doi:10.18653/v1/2023.acl-long.41 Hundt, A., Agnew, W., Zeng, V., Kacianka, S., & Gombolay, M. (2022). Robots enact malignant stereotypes. <i>Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FACCT '22)</i>, USA, 743–756. https://doi.org/10.1145/3531146.3533138 Koivisto, M., & Grassini, S. (2023). Best humans still outperform artificial intelligence in a creative divergent thinking task. <i>Scientific Reports</i>, 13, 13601. https://doi.org/10.1038/s41598-023-40858-3 Liang, W., Yuksekogonul, M., Mao, Y., Wu, E., & Zou, J. (2023). GPT detectors are biased against non-native English writers. <i>Patterns</i>, 4(7), 100779. https://doi.org/10.1016/j.patter.2023.100779 Marlow, R., & Wood, D. (2021). Ghost in the machine or monkey with a typewriter—generating titles for Christmas research articles in <i>The BMJ</i> using artificial intelligence: Observational study. <i>BMJ</i>, 375, e067732. doi:10.1136/bmj-2021-067732 Menz, B. D., Modi, N. D., Sorich, M. J., & Hopkins, A. M. (2023). Health disinformation use case highlighting the urgent need for artificial intelligence vigilance: Weapons of mass disinformation. <i>JAMA Internal Medicine</i>, November 13, 2023. doi:10.1001/jamainternmed.2023.5947 Miller, E. J., Steward, B. A., Witkower, Z., Sutherland, C. A. M., Krumhuber, E. G., & Dawel, A. (2023). AI hyperrealism: Why AI faces are perceived as more real than human ones. <i>Psychological Science</i>, 34(12), 1390-1403. https://doi.org/10.1177/09567976231207095 Peiris, H., Hayat, M., Chen, Z., Egan, G., & Harandi, M. (2023). Uncertainty-guided dual-views for semi-supervised volumetric medical image
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	segmentation. <i>Nature Machine Intelligence</i>, 5, 724–738. https://doi.org/10.1038/s42256-023-00682-w Peng, C., Yang, X., Chen, A., Smith, K. E., PourNejatian, N., Costa, A. B., Martin, C., Flores, M. G., Zhang, Y., Magoc, T., Lipori, G., Mitchell, D. A., Ospina, N. S., Ahmed, M. M., Hogan, W. R., Shenkman, E. A., Guo, Y., Bian, J., & Wu, Y. (2023). A study of generative large language model for medical research and healthcare. <i>NPJ Digital Medicine</i>, 6(210). https://doi.org/10.1038/s41746-023-00958-w Restrepo, D. D. J., Spagat, M., van Weezel, S., Zheng, M., & Johnson, N. F. (2020). A computational science approach to understanding human conflict. <i>Journal of Computational Science</i>, 46, 101088. https://doi.org/10.1016/j.jocs.2020.101088 Shea, Y., Lee, C. M. Y., Ip, W. C. T., Luk, D. W. A., Wong, S. S. W. (2023). Use of GPT-4 to analyze medical records of patients with extensive investigations and delayed diagnosis. <i>JAMA Network Open</i>, 6(8), e2325000. doi:10.1001/jamanetworkopen.2023.25000 Spapé, M., Davis, K. M., Kangassalo, L., Ravaja, N., Sovijärvi-Spapé, Z., & Ruotsalo, T. (2023). Brain-computer interface for generating personally attractive images. <i>IEEE Transactions on Affective Computing</i>, 14(1), 637-649. doi:10.1109/TAFFC.2021.3059043 Spitale, G., Biller-Andorno, N., & Germani, F. (2023). AI model GPT-3 (dis)informs us better than humans. <i>Science Advances</i>, 9, eadh1850. doi:10.1126/sciadv.adh1850 Sun, M., & Bardhan, R. (2023). Identifying hard-to-decarbonize houses from multi-source data in Cambridge, UK. <i>Sustainable Cities and Society</i>, 100, 105015. https://doi.org/10.1016/j.scs.2023.105015 Valeri, J. A., Soenksen, L. R., Collins, K. M., Ramesh, P., Cai, G., Powers, R., Angenent-Mari, N. M., Camacho, D. M., Wong, F., Lu, T. K., & Collins, J. J. (2023). BioAutoMATED: An end-to-end automated machine learning tool for explanation and design of biological sequences. <i>Cell Systems</i>, 14, 525-542. https://doi.org/10.1016/j.cels.2023.05.007
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